

NATIONAL CYBERSECURITY LAW

Design Project for VIT — Academic Assignment

Subject

Cybersecurity Law & Policy

Project Type

Group / Individual Project

Total Marks

100 Marks

Deliverables

Report + Presentation + Flowcharts +
Infographics

Important Notice to Students

This is a comprehensive project assignment. Read all sections carefully before starting. Your team must draft a complete Cybersecurity Law for the fictional country of VIT's. Each part of this project corresponds to a section of the law you will create. Be creative, thorough, and technically accurate.

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PART 1: Introduction

Prepare an overview section that sets the foundation for your cybersecurity law. This section should clearly articulate why cybersecurity legislation is necessary for VIT's.

What to Include

- Purpose of the law — What is this law designed to achieve?
- Importance of cybersecurity — Why does VIT's need this law now?
- Challenges faced by VIT's — Describe specific threats from the scenario
- Goals and expected outcomes — What will change after this law is enacted?

Purpose of the Law

As technology continues to play an important role in everyday life, citizens of VIT's increasingly depend on digital services for banking, education, healthcare, communication, and government services. While these technologies provide convenience and efficiency, they also create opportunities for cybercriminals to exploit vulnerabilities and target individuals and organizations.

The purpose of this Cybersecurity Law is to establish a legal framework that protects citizens, businesses, and government systems from cyber threats. The law aims to promote responsible use of technology, strengthen cybersecurity practices, and ensure that cybercrimes are investigated and punished effectively.

Importance of Cybersecurity

Cybersecurity is essential for maintaining trust in the digital economy and protecting sensitive information. Personal data, financial records, healthcare information, and government systems are increasingly stored and managed online. Without proper security measures, these systems may become targets for hackers, fraudsters, and other malicious actors.

Challenges Faced by VIT's

VIT's faces several cybersecurity challenges as its digital infrastructure continues to grow. These challenges include ransomware attacks on hospitals, online banking fraud, identity theft, phishing scams, cyberbullying, misinformation campaigns, and data breaches affecting both public and private organizations.

Also new technologies like IoT, AI and cloud computing introduce new security risks that require proper regulation.

Expected Outcomes

The implementation of this Cybersecurity Law is expected to create a safer digital environment for citizens, businesses, and government institutions. The law will strengthen data protection, improve cybercrime investigations, encourage responsible use of technology, and enhance national cybersecurity preparedness.

Scenario Context

VIT's is a rapidly growing digital nation with 50 million internet users, online banking, e-commerce platforms, government digital services, smart city infrastructure, AI-based public services, and cloud-based businesses. Recently, VIT's experienced large-scale data breaches, ransomware attacks on hospitals, banking fraud, social media misinformation campaigns, identity theft cases, and cyberbullying incidents.

PART 2: Definitions Section

Define at least 20 important cyber law terms. Your definitions must be precise, legally clear, and relevant to VIT's context. Below are the required terms — you may add more.

Required Terms to Define

#	Term	Definition (To Be Completed by Student)
1	Cybercrime	Cybercrime means any illegal activity carried out using a computer, network, digital device, or the internet that causes harm to individuals, organizations, or the government.
2	Hacker	A hacker is a person who gains access to computer systems or networks by identifying and exploiting security weaknesses.
3	Ethical Hacker	An ethical hacker is a cybersecurity professional who is authorized to test computer systems and identify security vulnerabilities for defensive purposes.
4	Malware	Malware refers to any software intentionally designed to damage, disrupt, monitor, or gain unauthorized access to computer systems and data.
5	Ransomware	Ransomware is a type of malware that blocks access to files or systems and demands payment in exchange for restoring access.
6	Phishing	Phishing is a fraudulent attempt to obtain sensitive information such as passwords, banking details, or personal information by pretending to be a trusted source.
7	Data Breach	A data breach occurs when personal, confidential, or sensitive information is accessed, disclosed, stolen, or exposed without authorization.
8	Personal Data	Personal data means any information that can directly or indirectly identify an individual, including names, addresses, phone numbers, email addresses, and identification numbers.
9	Sensitive Data	Sensitive data refers to information that requires a higher level of protection, such as financial records, medical information, biometric data, passwords, and government-issued identification details.
10	Digital Signature	A digital signature is an electronic method used to verify the identity of a person and confirm the authenticity of a digital document or transaction.
11	Electronic Evidence	Electronic evidence means any information stored or transmitted in digital form that may be used during an investigation or legal proceeding.

12	Identity Theft	Identity theft occurs when a person uses another individual's personal information without permission for fraudulent or illegal purposes.
13	Artificial Intelligence	Artificial Intelligence refers to computer systems or software capable of performing tasks that normally require human intelligence, such as learning, decision-making, and problem-solving.
14	Cloud Computing	Cloud computing is the delivery of computing services such as storage, software, and processing power over the internet instead of local devices.
15	Critical Infrastructure	Critical infrastructure includes systems and services that are essential for the functioning of society, such as banks, hospitals, transportation networks, power plants, and government systems.
16	Cyberbullying	Cyberbullying is the use of digital platforms to harass, threaten, insult, or intentionally harm another person.
17	Deepfake	A deepfake is digitally created or altered content, often generated using Artificial Intelligence, that falsely represents a person's appearance, voice, or actions.
18	Encryption	Encryption is the process of converting information into a secure format so that only authorized individuals can access or read it.
19	Firewall	A firewall is a security system that monitors and controls incoming and outgoing network traffic to protect devices and networks from unauthorized access.
20	Zero-Day Exploit	A zero-day exploit is a cyberattack that takes advantage of a software vulnerability before the software developer becomes aware of it or releases a fix.

PART 3: Citizens' Digital Rights

Create a comprehensive Bill of Digital Rights for every citizen of VIT's. For each right, provide a clear explanation of what it means and how it will be enforced.

Required Rights

Right to Privacy

Citizens must be protected from unauthorized collection, use, or sharing of their personal data.

This right means that nobody should collect personal data without permission. To collect personal data, there should be consent of the user and notices should be given. Organizations that collect personal data must clearly explain that how the data will be used and how long it will be there with them. Unnecessary data should not be collected.

Citizens may file complaints if their privacy has been violated. Companies found guilty may face penalties or suspension or ban.

Right to Secure Data

All organizations must store and protect citizens' data using industry-standard security measures.

This right means that organizations or companies must protect user information. Proper security measures should be taken. Data should be encrypted and the servers should be secure. Companies should use multi-factor authentication and secure backups.

If data is breached by a company fault, they must compensate the affected individual and face penalty.

Right to Data Deletion

Citizens may request the complete deletion of their personal data from any organization's systems.

People should be able to request for removal of their personal data. This right allows individuals to maintain control over their personal data. Organizations must provide a simple process through which citizens can submit deletion requests. Once a request is valid, company must remove the data from its system.

If organization refuses, it must provide a valid justification.

Right to Information

Citizens have the right to know what data is collected about them and how it is used.

There should be transparency because this is a fundamental right and this also helps in building trust between citizens and organizations. An organization collecting personal information must provide citizens with clear information regarding the purpose of collection and where it will be shared with, and the duration it will be there with.

If the organization fails to give proper valid reason then this may result to investigation and penalties.

Right to Report Cybercrime

Citizens can report cybercrimes without fear of retaliation, and authorities must act within defined timeframes.

Reporting cybercrimes should be simple and accessible to every citizen. There should be online portals and emergency helplines and also there will be a rule that if the citizen wants to report anonymously, they can do it. People should be able to report online frauds, cyberbullying also.

If some officials ignore this complaints of the citizen , then the action will be taken against the official.

Right to Fair Investigation

Any person accused of a cybercrime is entitled to a fair, transparent, and timely investigation.

A person should not be treated as guilty without proper proof. Digital evidence like chats, mails, screenshots must be carefully checked. Evidences should be collected legally. No one should be punished based only on suspicion.

If someone believes that their rights were violated during an investigation, they can file a complaint.

Instruction

You must include at least 6 rights. You may add more. Each right should be at least 2-3 paragraphs with specific implementation details.

PART 4: Cyber Offenses and Crimes

Define categories of cybercrime. For each offense, specify a description, real-world example within VIT's context, criminal punishment, and financial fine. Complete the table below.

Crime	Description + Example	Punishment	Fine
Unauthorized Access	Accessing a computer or data without permission. Example : If someone like a student accessed the government portal and saw the public data which is not meant to expose to public.	Up to 3 years imprisonment	Up to 5 Lakh Rupees
Data Theft	Stealing someone's personal data, their personal or business or work information from a system. Example : If someone copies the data like an employee and then sells it online.	Up to 4 years imprisonment	Up to 7 Lakh Rupees + Compensation
Identity Theft	Stealing someone's identity, using another personal identity for illegal things. Example : Someone uses somebody's personal id like aadhar card or pan card to open fake bank accounts.	Up to 5 years imprisonment	Up to 7 Lakh Rupees + Compensation
Phishing	Sending fake emails or messages and trick people to reveal their identity or bank information and also emotionally scamming them. Example : Somebody receives fake bank messages for kyc and then asking for bank id and passwords.	Up to 6 years imprisonment	Up to 10 Lakh Rupees
Ransomware	Encrypting important files of somebody's system and then demands money for accessing those files. Example : A hospital's patient records are encrypted and hackers demand payments in bitcoins to restore them.	Up to 12 years imprisonment	Up to 20 Lakh Rupees
Malware Distribution	Spreading harmful softwares like virus or keyloggers that damages and steals information. Example : A hacker shared a document whose title was important for the victim and when the file was opened it spread virus.	Up to 6 years imprisonment	Up to 8 Lakh Rupees
Cyberbullying	Digitally harassing people or stalking them or humiliating people online. Example : Student posting abusive content about another student on	Up to 3 years imprisonment	Up to 3 Lakh Rupees

	social media.		
Online Harassment	Online abuse, threats, or unwanted communication. Example : A person sending threatening messages to a victim through social media.	Up to 5 years imprisonment	Up to 5 Lakh Rupees
Financial Fraud	Making people fool online to steal their money. Example : Somebody asks for otp to the victim for their kyc or somebody asks for online payment to their qr code because mistakenly they have transferred the money to the victim.	Up to 7 years imprisonment	Up to 10 Lakh Rupees
Misinformation	Spreading false information that causes confusion or panic in society. Example : Making fake social media posts to create panic about college.	Up to 5 years imprisonment	Up to 5 Lakh Rupees
Website Defacement	Illegal modification of a website content. Example : Hackers replace the main page with some message of the website.	Up to 5 years imprisonment	Up to 7 Lakh Rupees
Cyber Terrorism	Conducting cyberattacks against a big infrastructure to create fear or against a nation for threat. Example : Attackers attack airport system or power grid systems.	Up to 20 years imprisonment	Up to 25 Lakh Rupees

PART 5: Data Protection Framework

Create a comprehensive data protection framework that all organizations operating in VIT's must follow. Your framework must cover the full data lifecycle.

Data Collection Rules

Organizations should only collect data which is genuinely required for their services. Before collecting any personal data, they should clearly explain where they are going to use that data and for how much period of time. They should not ask unnecessary data which is not needed for that service.

Companies must be transparent about their data collection practices. Information should be collected in a legal manner.

Data Storage Rules

Organizations should store the data safely by encrypting the data. Proper steps should be taken to keep personal data safe from hackers or data leaks. Security measures like encryption, firewalls and system updates should be used to protect data.

Access to private information should be limited to the administrative employees only. There should be regular monitoring and regular checkups of the servers and the software.

Data Sharing Rules

Organizations should not share personal data with other organizations unless the user has given permission. Citizens must be informed whenever their data is shared with third parties.

If data is shared with business partners or service providers, those parties must also follow the same security and privacy standards required under this law. Organizations remain responsible for protecting the data even after it has been shared.

User Consent Requirements

Organizations must obtain clear and informed consent from the users before collecting any personal information. Consent forms should be written in simple language so that users can easily understand.

Citizens should also have the option to withdraw their consent at any time. Once consent is withdrawn, organizations must stop processing the user's data unless there is a legal reason to continue.

Data Retention Period

Organizations should keep personal data only for as long as it is needed. Data should not be stored even if the purpose of storing data is completed.

Different types of data may have different retention periods. Once the required period has expired, organizations should securely delete the information

Data Deletion Process

Organizations should delete the data if the citizen requests. Organizations must provide an easy process through which users can submit deletion requests.

After receiving a valid request, the organization should remove the data within a reasonable time frame and inform the user once the deletion has been completed. If the request cannot be fulfilled, a clear explanation must be provided.

Breach Notification Requirements

If a data breach occurs, the affected organization must immediately investigate the incident and report it to the Authority within 72 hours. Quick reporting helps reduce damage and allows authorities to provide assistance.

Organizations must also notify affected users if their personal information may have been exposed. They should explain users like what happened and what was affected and what steps users can take to protect themselves.

PART 6: Corporate Responsibilities

All companies and organizations operating in VIT's must comply with the following cybersecurity responsibilities. Create at least 10 specific, enforceable corporate obligations.

Required Responsibilities (Minimum 10)

Responsibility 1: Data Protection Measures

Every company that collects customer information must take reasonable steps to keep that information safe. This includes using secure passwords, encryption, and limiting access to authorized employees only.

This rule applies to all businesses that handle personal data. If a company fails to protect customer information and a breach occurs due to negligence, it may face fines and further investigation by the National Cybersecurity Authority.

Responsibility 2: Incident Reporting

If a company experiences a cyberattack or data breach, it must inform the NCSA as soon as possible and no later than 72 hours after discovering the incident.

If a company tries to hide a cyberattack or delay the report, it can make the situation worst. Companies that fail to report incidents on time may face fines or legal actions.

Responsibility 3: Employee Cybersecurity Training

Companies must regularly train employees on basic cybersecurity practices. Employees should know how to identify suspicious emails, create strong passwords, and handle sensitive information safely.

Since human mistakes are one of the main causes of cyber incidents, organizations that ignore employee training requirements may be penalized.

Responsibility 4: Security Audits

Every organization should regularly check its systems for weaknesses before cybercriminals can exploit them. Security audits help companies identify problems and fix them early. Companies that fail to perform regular audits may face legal actions.

Responsibility 5: Appointment of Data Protection Officer (DPO)

Large organizations that handle a lot of customer information must appoint a Data Protection Officer. This person will ensure that data is collected, stored, and used according to the law.

If a company fails to appoint a responsible officer despite handling sensitive information, it may face regulatory action.

Responsibility 6: Access Control Management

Not every employee needs access to all company information. Organizations must make sure employees can only access the data required for their jobs.

Poor access control increases the risk of data leaks and insider threats. Violations may lead to penalties and mandatory security improvements.

Responsibility 7: Encryption of Sensitive Data

Important information such as banking details, medical records, and identity documents should always be encrypted. Encryption makes data difficult to read even if attackers manage to steal it.

Organizations that fail to encrypt sensitive information may be held responsible for resulting damages.

Responsibility 8: Backup and Recovery Systems

Companies should keep regular backups of important data and have a recovery plan in case of ransomware attacks or system failures.

Without proper backups, businesses may lose valuable information permanently. Failure to maintain backup systems may result in penalties or legal actions.

Responsibility 9: Penetration Testing

Companies should hire cybersecurity experts to test their systems, to do penetration testing and find vulnerabilities and identify weaknesses before attackers can exploit them.

Findings must be documented and actions should be taken.

Ignoring known vulnerabilities may result in fines.

Responsibility 10: Incident Response Plan

Every company should have a clear plan for dealing with cyberattacks. Employees should know what steps to take when an incident occurs, who should be informed, and how systems will be restored.

The plan shall be reviewed and updated annually.

Organizations that fail to prepare for cyber incidents may face greater penalties if poor planning worsens the damage.

Examples to Consider

Maintain security controls | Conduct regular security audits | Train employees on cybersecurity | Report incidents within 72 hours | Protect customer information | Appoint a Data Protection Officer | Conduct annual penetration testing | Implement access control policies | Encrypt sensitive data | Maintain incident response plans

PART 7: Government Responsibilities

Specify how government agencies of VIT's must protect national interests in cyberspace. Include concrete mechanisms for each area.

Protect National Infrastructure

The government of VIT's will identify and protect critical infrastructure such as banks, hospitals, airports, railways, power plants, and government data centers. These systems are essential for the daily functioning of the country, and any cyberattack on them could have serious consequences for citizens.

To strengthen security, organizations must follow strict rules, strict cybersecurity standards, conduct regular audits, maintain backups and immediately report major cyber incidents to the National Cyber Security Authority.

Monitor Cyber Threats

The government will continuously monitor cyber threats that may affect citizens, businesses, and public services. Cybercriminals constantly develop new attack methods, making it necessary to detect threats before they cause significant damage.

The National Cyber Security Authority will operate a national threat monitoring center responsible for collecting threat intelligence, tracking cyberattack trends, and issuing alerts whenever new risks are identified. Information gathered through monitoring activities shall be used only for cybersecurity purposes and in accordance with privacy laws.

Investigate Cybercrimes

The government will establish specialized cybercrime investigation units with trained officers, digital forensic experts, and cybersecurity professionals. These teams will be responsible for investigating cyber offenses such as hacking, identity theft, financial fraud, ransomware attacks, and data breaches.

Investigations must follow legal procedures and respect the rights of all individuals involved. Digital evidence must be properly collected and preserved so that it can be used in court. The National Cyber Security Authority may work together with law enforcement agencies and other government departments during complex investigations.

International Coordination

Cybercrime often crosses national borders, making international cooperation essential. The government of VIT's shall work with foreign governments, international organizations, and cybersecurity agencies to investigate cross-border cybercrimes and share information about emerging threats.

The National Cyber Security Authority may participate in international cybersecurity programs, threat intelligence exchanges, and joint investigations. Such cooperation will help VIT's respond more effectively to cyberattacks.

Awareness Programs

The government will promote cybersecurity awareness among citizens through educational campaigns, workshops, school programs, and social media initiatives. Many cyberattacks succeed because people are unaware of common online threats such as phishing scams, weak passwords, and fake websites.

The National Cyber Security Authority will regularly publish cybersecurity guidelines and conduct awareness campaigns to educate the public about safe online practices. Special programs may be organized for students, senior citizens, and small businesses, as these groups are often targeted by cybercriminals.

PART 8: Cyber Incident Reporting Framework

Design a step-by-step process for how cyber incidents are reported, investigated, and resolved in VIT's. You must also create a flowchart (as a separate deliverable).

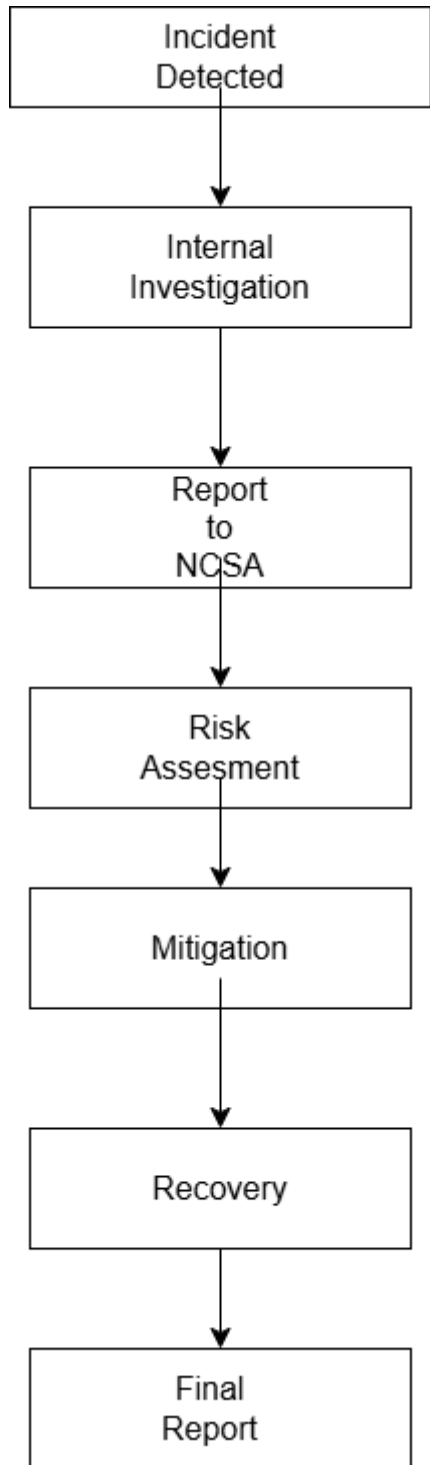
Reporting Process Steps

Step	Stage	Description (Student to Complete)
1	Incident Detected	The process begins when a cyber incident is discovered. This could be a ransomware attack, a data breach, suspicious login activity, or any other security issue. As soon as the incident is detected, the organization should secure the affected systems and prevent the situation from getting worse. Basic information about the incident should be recorded, including when it was discovered, which systems were affected, and who reported it. Early action is important because it helps reduce damage and preserve evidence.
2	Internal Investigation	After the incident is detected, the organization's cybersecurity team should investigate what happened. They should examine system logs, user activity, and security alerts to understand how the attack occurred. The goal of this stage is to identify the source of the problem, estimate the damage, and determine whether sensitive information has been affected. All findings should be documented carefully.
3	Report to Authority	If the incident is serious, the organization must report it to the National Cyber Security Authority (NCSA) within 72 hours. The report should explain what happened, what systems were affected, and what actions have already been taken. Reporting incidents quickly allows the government to provide support and helps prevent similar attacks from affecting other organizations.
4	Risk Assessment	Once the incident has been reported, the organization and the NCSA will assess how serious the situation is. They will examine factors such as the amount of data affected, financial losses, disruption of services, and possible risks to citizens. Based on this assessment, the incident will be categorized as low, medium, or high risk. This helps determine the level of response required.
5	Mitigation	At this stage, efforts are focused on stopping the attack and reducing further damage. The organization may disconnect infected systems, remove malicious software, block suspicious

		accounts, and strengthen security controls. The main objective is to contain the incident as quickly as possible while protecting important data and services.
6	Recovery	After the threat has been removed, the organization can begin restoring affected systems and services. Data may be recovered from backups, and systems should be tested to ensure they are functioning properly before normal operations resume. Organizations should also review their security practices and implement improvements to prevent similar incidents in the future.
7	Final Report	Once the incident has been resolved, a final report should be prepared and submitted to the NCSA. The report should describe what happened, how the issue was handled, the damage caused, and the lessons learned from the incident. This report helps improve future responses and allows organizations to strengthen their cybersecurity measures based on real-world experience.

Deliverable Reminder

You must submit a separate FLOWCHART showing this complete process. Use tools like Canva, draw.io, Lucidchart, or PowerPoint SmartArt to create a professional diagram.



PART 9: National Cyber Security Authority (NCSA)

Design the organizational structure, responsibilities, and powers of the NCSA — the primary government agency responsible for cybersecurity in VIT's.

Organizational Structure

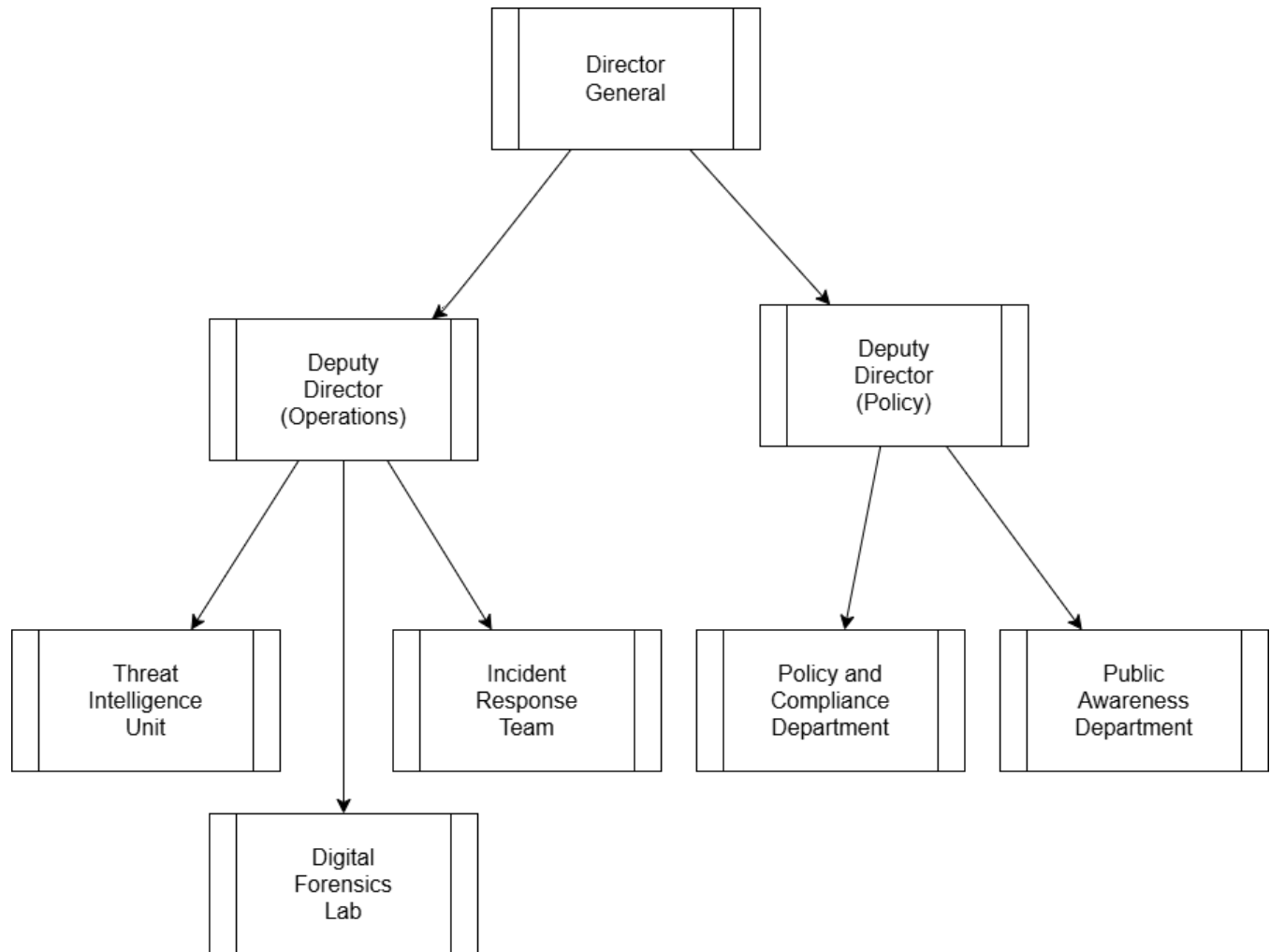
The National Cyber Security Authority (NCSA) will be the main government agency responsible for cybersecurity in VIT's. Its primary role is to protect citizens, organizations, and critical infrastructure from cyber threats while ensuring that cybersecurity laws are properly enforced.

The NCSA will be headed by a Director General appointed by the government for a fixed five-year term. The Director General will be supported by two Deputy Directors. One Deputy Director will see about the cyber incident reports and investigations, while the other will manage policy making, monitoring, international relations and public awareness programs.

All departments within the NCSA will follow a clear reporting structure. Department heads will report to their respective Deputy Directors, who will then report to the Director General. This structure is most efficient for faster communication and decision making.

STRUCTURE

1. Director General
2. Deputy Directors
3. Department Heads
4. Cybersecurity Experts
5. Investigators and Analysts



Core Responsibilities

The main responsibility of the NCSA is to protect VIT's digital environment and reduce cyber risks before they become major incidents. The agency will develop national cybersecurity policies, recommend updates to cybersecurity laws, and provide technical advice to the government on emerging technologies such as Artificial Intelligence, Cloud Computing, and Quantum Computing.

The NCSA will operate for 24/7 and 24/7 monitoring will be done that continuously monitors cyber threats affecting banks, hospitals, government systems and transportation systems and critical infrastructure. When a major attack occurs, the NCSA will response to organizations which are affected and also response law enforcement agencies.

The authority will also conduct cybersecurity audits, certify organizations that meet security standards, and run public awareness campaigns such as "Cybersafe VIT's" to educate citizens about online safety, phishing attacks, identity theft, and digital privacy.

Legal Powers

To effectively protect the country from cyber threats, the NCSA will be granted specific legal powers under the Cybersecurity Law of VIT's.

The authority may request information related to cybercrime investigations from internet service providers, cloud service providers, social media companies, and other organizations. Such requests must follow legal procedures and be used only for authorized investigations. Organizations receiving these requests must cooperate within the time period specified by law.

The NCSA may also issue Cyber Takedown notices for websites, domains or online services involved in cybercrime activities such as phishing, malware distribution, ransomware operations etc. In urgent situations involving national security or critical infrastructure, the NCSA may temporarily suspend harmful online services.

The authority will also have the power to impose financial penalties on organizations, issue security improvement notices or issue enquiry about some security purpose.

Investigation Process

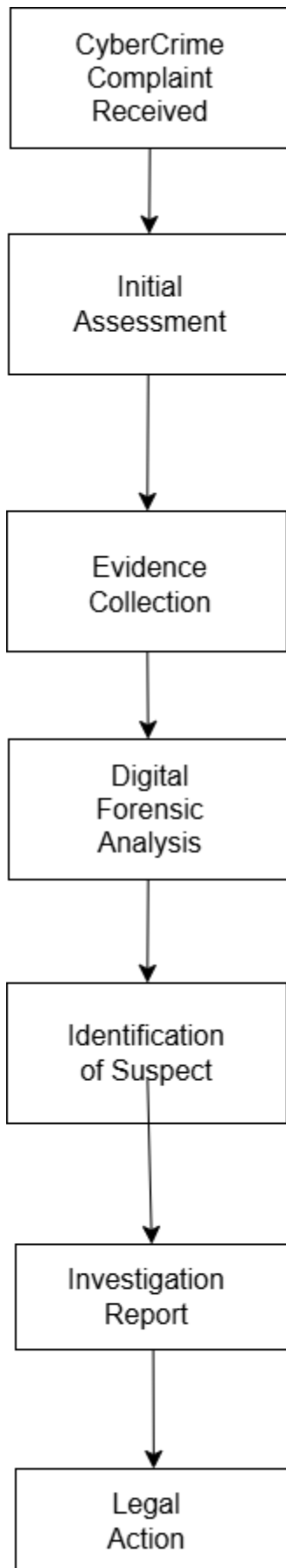
Whenever a cybercrime complaint is received, the NCSA will follow a structured investigation process to ensure that evidence is handled properly and investigations are conducted fairly.

Complaints may be submitted through an online reporting portal, a cybersecurity helpline, or local cybercrime reporting centers. After receiving a complaint, the NCSA will assign a case reference number and perform an initial assessment to determine the risk of incident and also the incident.

Digital evidence such as logs, emails, screenshots, and device records will then be collected and preserved. Digital forensic experts will analyze the evidence using approved forensic tools to identify attack methods, trace suspects, and determine the extent of the damage. If the investigation identifies criminal activity, the case will be forwarded to law enforcement agencies and Cyber Courts for further legal action.

Investigation Steps

1. Complaint Received
2. Initial Assessment
3. Evidence Preservation
4. Digital Forensic Analysis
5. Identification of Suspects
6. Submission of Findings
7. Legal Action and Court Proceedings



Departments

To handle different cybersecurity challenges, the NCSA will consist of several specialized departments.

1. Threat Intelligence Unit (TIU)

This department will monitor global cyber threats, identify emerging attack trends, and share threat intelligence with critical sectors such as banks, hospitals, and government agencies. Its goal is to detect threats early and help organizations prepare for potential attacks.

2. Incident Response Team (IRT)

The Incident Response Team will respond to major cybersecurity incidents such as ransomware attacks, large-scale data breaches, and attacks on critical infrastructure. The team will work to contain incidents quickly and restore normal operations.

3. Digital Forensics Laboratory (DFL)

The Digital Forensics Laboratory will collect, preserve, and analyze digital evidence used in cybercrime investigations. The department will ensure that evidence remains reliable and can be presented in court when necessary.

4. Policy and Compliance Department

This department will monitor compliance with cybersecurity laws and regulations. It will conduct audits, evaluate organizational security practices, and recommend improvements where necessary.

5. Public Awareness and Education Department

This department will lead cybersecurity awareness campaigns across schools, colleges, businesses, and communities. It will educate citizens about safe online behavior, phishing scams, privacy protection, and cybercrime reporting procedures.

PART 10: AI and Emerging Technology Regulations

Create specific cybersecurity regulations for emerging technologies used in VIT's. For each technology, identify key risks and write enforceable rules.

Artificial Intelligence

Key Risks

Artificial Intelligence is becoming a part of many important services such as healthcare, banking, hiring, and security systems. If these systems are not properly secured, hackers may manipulate the data used by AI models and cause incorrect decisions. AI can also be used by cybercriminals to create realistic phishing emails, fake content, or automated cyberattacks.

Another concern is that people may depend too much on AI systems and ignore human judgment. If an AI system makes a mistake, it could affect a large number of people.

Regulations

Any AI system used in important sectors such as healthcare, banking, law enforcement, or recruitment must be registered with the National Cyber Security Authority (NCSA) before deployment.

Organizations using AI must conduct regular security and performance audits to ensure that the system is operating safely and fairly. Citizens affected by important AI decisions must have the right to request a human review. Organizations will remain responsible for decisions made by their AI systems.

Deepfakes

Key Risks

Deepfakes can be used to create fake videos, images, or audio recordings that appear real. Such content can be used for fraud, spreading misinformation, damaging reputations, or influencing public opinion during elections.

Deepfakes may also be used for online harassment and identity theft by impersonating other individuals.

Regulations

All AI-generated media shared publicly must be clearly labeled as AI-generated content whenever possible.

Creating or sharing deepfakes for fraud, harassment, defamation, or election interference shall be treated as a criminal offense. Social media platforms operating in VIT's must take reasonable steps to identify and remove harmful deepfake content after receiving verified complaints.

IoT Devices

Key Risks

Internet of Things (IoT) devices such as smart cameras, smart meters, sensors, and connected appliances are often connected to the internet for long periods of time. Weak security settings or outdated software can make these devices easy targets for hackers.

A compromised IoT device may allow attackers to access larger networks or disrupt important services.

Regulations

Manufacturers must ensure that IoT devices meet basic security standards before being sold or deployed. Devices should not use weak default passwords, and users must be allowed to change security settings.

Manufacturers must provide security updates for a reasonable period or a timed period after the device is released. Government and smart city IoT devices must be placed on secure networks separated from public internet services whenever possible.

Cloud Computing

Key Risks

Many organizations store their data or important information in cloud platforms. Poor security settings, weak access controls, and data breaches can expose sensitive information. Another concern is that critical national data may be stored in foreign locations with different legal protections.

A major problem at a cloud provider could also affect multiple services at the same time.

Regulations

Cloud service providers handling government or critical infrastructure data must follow

security standards approved by the NCSA.

Sensitive government information should be stored within VIT's or in locations that provide equivalent levels of data protection. Cloud providers must report security breaches and ensure that data is encrypted during storage and transmission.

Digital Identity Systems

Key Risks

Digital identity systems store important personal information such as identification details and biometric records. If this information is stolen, citizens may face identity theft and long-term privacy risks.

Overdependence on digital identity systems may also create difficulties for citizens who have limited access to digital technology.

Regulations

Personal and biometric information must be encrypted and protected using strong security controls. Information collected for one purpose should not be used for another purpose without the citizen's consent.

Citizens who are unable to use or don't want to use digital identity services must be provided with alternative methods to access government services. Unauthorized access to national identity databases will result in severe penalties and legal action.

Autonomous Systems

Key Risks

Autonomous systems such as drones, self-driving vehicles and robotic machines depend on software and communication systems. If hackers gain control of these systems, they may cause physical damage, financial losses and also risks to public safety.

Software failure and communication failure may also affect the autonomous systems.

Regulations

All autonomous systems used by government agencies or businesses must undergo cybersecurity testing before deployment. Communication between devices and control systems must be encrypted to prevent unauthorized access.

Manufacturers will be responsible for security flaws in the design of the system, while operators will be responsible for improper use. Drones above a specified size or weight must be registered and operated according to safety regulations.

PART 11: Critical Infrastructure Protection

Create sector-specific cybersecurity requirements for each of the following critical infrastructure types in VIT's. Requirements should be practical and technically grounded.

Sector	Cybersecurity Requirements (Student to Complete)
Banks & Financial	Banks handle large amounts of money and sensitive customer information, making them a common target for cybercriminals. To protect customers, banks should use multi-factor authentication for online banking and important transactions. Security checks should be carried out regularly to identify weaknesses before attackers can exploit them. Banks should also monitor suspicious transactions and unusual account activity in real time. Important banking systems should be kept separate from customer-facing websites and mobile applications to reduce the risk of large-scale attacks. Regular backups and recovery plans should also be maintained so that services can continue during emergencies.
Hospitals & Healthcare	Hospitals store sensitive patient information and depend heavily on digital systems for treatment and medical records. A cyberattack on a hospital can affect both data security and patient safety. Patient records should be encrypted and accessible only to authorized medical staff. Hospitals must regularly back up important data and maintain emergency procedures so that doctors and nurses can continue providing care even if computer systems become unavailable. Medical equipment and patient care systems should be protected from unauthorized access and monitored regularly.
Airports & Aviation	Airports rely on digital systems for flight operations, passenger management, and communication. If these systems are compromised, it can cause delays, confusion, and safety concerns. Access to critical airport systems should be limited to authorized personnel. Regular security assessments should be conducted to identify vulnerabilities and improve protection measures. Airports should also maintain backup communication systems to ensure operations can continue during cyber incidents.
Railways & Transport	Modern railway systems depend on technology for train scheduling, ticketing, signaling, and communication. Any disruption to these systems can affect thousands of passengers.

	Railway control systems should be protected from unauthorized access and continuously monitored for suspicious activity. Ticketing systems and public Wi-Fi services should be separated from critical operational systems to reduce security risks. Regular security audits should also be conducted to ensure safe and reliable operations.
Power Plants & Energy	Power plants and energy providers are part of the country's critical infrastructure. A successful cyberattack on these systems could affect entire cities and disrupt essential services. Access to power management systems should be restricted to authorized employees only. Security monitoring tools should be used to detect unusual activity, and backup control systems should be available if primary systems fail. Regular cybersecurity drills should also be conducted to prepare staff for potential incidents.
Government Data Centers	Government data centers store important information related to citizens, public services, and government operations. Protecting this information is essential for national security and public trust. Strong security measures such as encryption, access controls, and continuous monitoring should be used to protect government data. Regular audits should be conducted to ensure compliance with cybersecurity standards. Backup systems and disaster recovery plans should also be maintained to ensure that essential government services remain available during cyber emergencies.

PART 12: Awareness and Education

Design a National Cybersecurity Awareness Campaign for VIT's. Your campaign must reach all segments of society through age-appropriate and effective programs.

School Programs (Age 6-18)

Cybersecurity education should start at an early age because children are spending more time online than ever before. For students between 6 and 10 years old, the focus should be on basic online safety such as keeping passwords secret, avoiding suspicious links, and understanding that personal information should not be shared with strangers online.

Students aged 11 to 14 should learn how to identify phishing messages, deal with cyberbullying, and use social media responsibly. For students between 15 and 18 years old, schools can introduce topics such as digital privacy, ethical hacking, cybersecurity careers, and the importance of protecting personal data. Every school should organize an annual Cyber Safety Day where students can participate in quizzes, awareness activities, and practical exercises.

College & University Programs

All higher education institutions shall provide cybersecurity awareness training to students. Students from any course should learn about basic cybersecurity concepts. Cybersecurity is important for every student, not just those studying computer science. Courses covering digital rights, cyber hygiene, and online safety shall be encouraged.

Universities offering technology-related programs shall provide specialized cybersecurity modules, practical laboratory sessions, internships, and participation in cybersecurity competitions.

Corporate Training Programs

Organisations shall provide cybersecurity awareness training to employees during onboarding and conduct annual refresher programs. Employees handling sensitive information shall receive additional specialized training. Employees are often the first target of phishing attacks and online scams. For this reason, every employee should receive cybersecurity training when joining an organization.

Training records shall be maintained and made available during compliance audits conducted by the NCSA.

Social Media Campaigns

The government shall use social media platforms to promote cybersecurity awareness among citizens. Educational content may include videos, posters, infographics, and public service announcements covering phishing, online fraud, privacy protection, and cybercrime reporting.

Campaigns shall be designed for different age groups to ensure effective communication. Campaigns can include short videos, reels, posters, quizzes, and live sessions on topics such as phishing scams, online fraud, password security, deepfakes, and privacy protection.

Public Awareness Workshops

The NCSA should organize regular cybersecurity awareness workshops across the country. These workshops can be held in schools, colleges, community centers, libraries, and local government offices.

Special attention should be given to senior citizens, who are often targeted by online fraud, and small business owners, who may not have dedicated cybersecurity teams. The workshops should focus on practical topics such as identifying scams, protecting personal information, securing online banking accounts, and reporting cybercrime incidents. Participation should be free so that everyone can benefit from these programs.

PART 13: International Cooperation

Define how VIT's will collaborate with other countries and international organizations to combat cybercrime that crosses national borders.

Cybercrime Investigations

Cybercrime often does not stop at national borders. A hacker sitting in one country can easily target victims in another country. Because of this, VIT's must work closely with other governments and international agencies when investigating serious cybercrimes. The National Cyber Security Authority (NCSA) will cooperate with foreign law enforcement agencies to collect evidence, track cybercriminals, and support investigations involving multiple countries. This cooperation can help identify attackers more quickly and improve the chances of successful prosecution.

Information Sharing Agreements

Cyber threats change every day, and no country can fight them alone. VIT's should work with international cybersecurity organizations and Computer Emergency Response Teams to share information about new cyber threats and vulnerabilities. Sharing information about malware, phishing campaigns, and hacking techniques can help organizations prepare for attacks before they cause significant damage. All information sharing must respect privacy laws and protect citizens' personal data.

Joint Operations

Some cybercriminal groups operate across several countries, making them difficult to stop through the efforts of a single nation. In such situations, VIT's may work together with other countries to investigate and disrupt these criminal networks. The NCSA may also participate in international cybersecurity exercises and training programs to improve its ability to respond to large-scale cyber incidents. These activities help strengthen cooperation and build trust between countries.

Extradition Requests

In some cases, cybercriminals may commit offenses against citizens of VIT's while living in another country. When this happens, the government may request help from international agreements and legal procedures.

Requests should only be made when there is sufficient evidence linking the suspect to the crime. This process helps ensure that cybercriminals cannot avoid responsibility simply by operating from another country.

Threat Intelligence Exchange

The NCSA should regularly exchange cyber threat information with international partners. If another country detects a new ransomware attack, malware campaign, or phishing technique, that information can be shared with VIT's before the threat spreads further. This early warning system allows organizations, government agencies, and critical infrastructure operators to strengthen their defenses and respond more effectively to emerging threats.

PART 14: Case Studies

Create 3 original fictional cybercrime cases set in VIT's. Each case must be realistic, detailed, and demonstrate how your law would be applied.

Case Study 1: Government Database Breach

Crime Category	Unauthorized Access / Data Breach
What Happened	A government department responsible for citizen services was using an old online portal that had not received security updates for a long time. A hacker discovered a weakness in the system and managed to gain access to the database. The attacker stole personal information belonging to nearly 50,000 citizens, including names, addresses, and identification details. A few weeks later, this information appeared for sale on an illegal website, raising concerns about identity theft and fraud.
Law Violated	Laws violated are : 1. Unauthorized Access 2. Data Theft 3. Violation of Privacy Rights
Investigation Process	After receiving complaints from affected citizens, the NCSA launched an investigation. Experts examined server logs, tracked suspicious activity, and identified the method used by the attacker. With the help of internet service providers and law enforcement agencies, investigators traced the hacker and collected digital evidence from his devices.
Punishment Given	The offender was found guilty of Unauthorized Access and Data Theft under the Cybersecurity Law of VIT's. The court sentenced him to 4 years imprisonment and imposed a fine of 7 lakh rupees.
Lessons Learned	After the incident, government departments were instructed to regularly update their systems and conduct security audits to prevent similar attacks.

Case Study 2: Fake Banking Website Scam

Crime Category	Financial Fraud / Phishing
What Happened	A group of cybercriminals created a fake website that looked exactly like the official website of VIT National Bank. They also sent thousands of fake SMS messages asking customers to verify their accounts. Many users entered their usernames, passwords, and OTPs on the fake website. The criminals then used this information to access bank accounts and transfer money to different accounts under their control.
Law Violated	Laws violated are : 1. Phishing 2. Identity Theft 3. Financial Fraud
Investigation Process	The NCSA received multiple complaints from bank customers who reported unauthorized transactions. Investigators analyzed the fake website, domain registration records, and banking transaction logs. The investigation revealed a network of individuals operating the scam from different cities. Law enforcement agencies coordinated arrests and seized several computers and mobile devices.
Punishment Given	The main offenders were convicted of Phishing, Financial Fraud, and Identity Theft. The court sentenced them to 7 years imprisonment and imposed a fine of 10 lakh rupees.
Lessons Learned	Banks introduced stronger customer awareness programs and improved phishing detection systems. Citizens were also educated about verifying website addresses before entering sensitive information.

Case Study 3: Ransomware Attack on Hospital

Crime Category	Ransomware / Cyber Terrorism
What Happened	A major hospital in VIT's became the victim of a ransomware attack after an employee accidentally opened a malicious email attachment. The malware quickly spread across the hospital network and encrypted patient records, appointment systems, and internal files. The attackers demanded a large ransom payment in exchange for restoring access to the data. As a result, hospital operations were disrupted, and several services had to be temporarily provided manually.
Law Violated	Laws violated are : 1. Ransomware Attack 2. Malware Distribution 3. Unauthorized Access 4. Cyber Terrorism
Investigation Process	The hospital immediately reported the incident to the NCSA. Cybersecurity specialists isolated infected systems and began forensic analysis to determine the source of the attack. Investigators identified the ransomware group through digital evidence and international cooperation with foreign cybersecurity agencies. The individuals responsible were eventually located and arrested.

Punishment Given	The offenders were convicted of Ransomware, Malware Distribution, and Cyber Terrorism. Due to the serious impact on public safety and healthcare services, the court sentenced the primary offenders to 20 years imprisonment and imposed a fine of 25 lakh rupees.
Lessons Learned	The incident demonstrated the importance of employee cybersecurity training, regular data backups, and network segmentation. Hospitals across VIT's were later required to strengthen their cybersecurity measures and conduct emergency response exercises.

PART 15: Future Recommendations

Based on your research, suggest forward-looking improvements and additions to VIT's cybersecurity law for the next 10 years. Consider how technology will evolve and what new threats will emerge.

AI Security (2025-2035)

Artificial Intelligence is developing very quickly and will become a bigger part of daily life in the coming years. While AI can improve efficiency and security, it can also be misused for cyberattacks, fake content creation, and large-scale phishing campaigns.

To address these risks, VIT's should regularly update its AI regulations and ensure that important decisions involving healthcare, banking, education, and law enforcement continue to involve human supervision. Organizations using AI should also be responsible for ensuring that their systems are safe, secure, and fair.

Quantum Computing

Even quantum computers are not widely used, they may eventually become powerful enough to break encryption used today. This could create major cybersecurity challenges in the future.

To prepare for this possibility, VIT's should invest in research and gradually adopt stronger encryption technologies. Government agencies, banks, and critical infrastructure operators should begin planning for future upgrades to their security systems.

Privacy Protection

As more devices collect personal information, protecting privacy will become increasingly important. Smart devices, wearable technology, and online services gather large amounts of data about people's activities and behaviour.

Future privacy laws should provide stronger protection for personal information and give citizens greater control over how their data is collected, stored, and shared. Organizations should also be more transparent about how they use customer information.

Digital Identity

Digital identity systems can make it easier for citizens to access online services, but they also create security risks if personal information is compromised. A large-scale identity breach could affect millions of people.

To reduce these risks, VIT's should continue improving the security of digital identity systems and ensure that strong encryption and authentication methods are used. Citizens

who are unable to access digital services should still have alternative options available.

Cyber Warfare

Cyberattacks against governments, power grids, hospitals, and communication networks are becoming more common around the world. In the future, cyber warfare may become as important as traditional warfare.

VIT's should strengthen its national cyber defenses, improve cooperation with other countries, and regularly conduct cybersecurity drills to prepare for large-scale attacks. Critical infrastructure should receive special protection because disruptions to these systems could affect millions of citizens.

Evaluation Criteria

Your project will be graded based on the following criteria. Review this carefully to maximize your marks.

Criteria	Max Marks	Marks Obtained
Research Quality — depth and accuracy of information, use of real-world examples	20	
Law Structure — completeness, logical organization, and legal clarity of the law	20	
Innovation — originality of solutions, creative approach to cybersecurity challenges	15	
Technical Accuracy — correctness of technical details, cybersecurity concepts, and procedures	15	
Presentation — quality of the PowerPoint presentation and oral delivery	10	
Diagrams & Flowcharts — clarity, accuracy, and professionalism of visual deliverables	10	
Case Studies — realism, detail, and correct application of the law	10	
TOTAL	100	

Deliverables Checklist

Before submitting, ensure you have completed all required deliverables:

Done	Deliverable	Requirements
☐	Project Report	Minimum 25 pages, well-formatted, all 15 parts completed, diagrams included
☐	PowerPoint Presentation	15 to 20 slides, covering all major sections of the law
☐	Flowchart: Cybercrime Reporting	Shows complete incident reporting process from detection to resolution
☐	Flowchart: Investigation Process	Shows NCSA investigation steps from complaint to judgment

Done	Deliverable	Requirements
☐	Flowchart: Data Breach Response	Shows data breach response procedure for organizations
☐	Infographic: Citizens' Rights	Visual summary of all citizens' digital rights
☐	Infographic: Cybercrime Categories	Visual taxonomy of all defined cybercrimes with penalties
☐	Infographic: Security Framework	Visual overview of the data protection and corporate security framework

Good Luck!

This project challenges you to think like a lawmaker, a technologist, and a citizen. Research real-world cybersecurity laws (India's IT Act, GDPR, Budapest Convention) for inspiration. Be creative, be thorough, and make VIT's digital future secure!